

AWS Provisioner

Setup & Installation Guide

Smart AWS Infrastructure Provisioning System

macOS | Linux | Windows

This guide covers everything needed to set up and run the complete application from scratch on any machine.

Quick Start (3 Commands)

1. Unzip the project
2. Run `setup.sh` (or `setup.bat` on Windows)
3. Run `start.sh` (or `start.bat` on Windows)

Open `http://localhost:3000` | Login: *Eternal-prithivi*

1 PREREQUISITES

Before running the setup, your machine needs the following software installed:

Required (Must Have)

Tool	Min Version	Check Command	macOS Install	Windows Install
Python	3.10+	python3 --version	brew install python	python.org/downloads
Node.js	18+	node --version	brew install node	nodejs.org (LTS)
npm	9+	npm --version	Comes with Node	Comes with Node

Optional (Web UI works without these)

Tool	Purpose	macOS	Windows
Terraform	Deploy infrastructure	brew install terraform	terraform.io/downloads
AWS CLI	Cloud operations	brew install awscli	aws.amazon.com/cli
OPA	Rego policy checks	brew install opa	github.com/open-policy-agent
Infracost	Cost estimation	brew install infracost	infracost.io

IMPORTANT - Windows Python Install

When installing Python on Windows, you MUST check the box 'Add Python to PATH' at the bottom of the installer screen. Without this, the 'python' command won't work.

2 macOS / LINUX SETUP

Automated Setup (Recommended)

Open Terminal, navigate to the project folder, and run:

```
chmod +x setup.sh
./setup.sh
```

This will automatically: check prerequisites, create a Python virtual environment, install all Python and Node.js dependencies, run 308 tests, and verify the build.

Start the Application

Option A - One command:

```
./start.sh          # Starts both servers
./start.sh --stop    # Stops both servers
```

Option B - Two terminals:

Terminal 1 (Backend API):

```
cd web-ui/api
../../.venv/bin/uvicorn server:app --reload --port 8000
```

Terminal 2 (Frontend):

```
cd web-ui/frontend
npm run dev
```

Then open <http://localhost:3000> in your browser.

IMPORTANT - uvicorn[standard]

You MUST install uvicorn[standard] (not just uvicorn). The [standard] extra adds WebSocket support needed for the CloudShell terminal. setup.sh does this automatically.

3 WINDOWS SETUP (COMPLETE GUIDE)

Step 1: Install Prerequisites

- 1) Download Python 3.10+ from python.org/downloads. During install, CHECK 'Add Python to PATH'. Click 'Install Now'.
- 2) Download Node.js LTS from nodejs.org. Run the installer with default settings.
- 3) Verify both: Open Command Prompt and run 'python --version' and 'node --version'.

Step 2: Run Setup

Option A (Easiest): In File Explorer, navigate to the project folder and double-click setup.bat. Wait ~2-3 minutes.

Option B (Command Prompt):

```
cd "C:\path\to\aws using terraform"  
setup.bat
```

Step 3: Start the Application

Option A: Double-click start.bat. Two Command Prompt windows open (one for backend, one for frontend) and your browser opens to localhost:3000.

Option B (Manual):

Window 1 - Backend:

```
cd web-ui\api  
..\..\venv\Scripts\uvicorn.exe server:app --reload --port 8000
```

Window 2 - Frontend:

```
cd web-ui\frontend  
npm run dev
```

Step 4: Stop the Servers

- Close the two Command Prompt windows that opened, OR
- Run: start.bat --stop

Kill Stuck Ports (Windows)

```
netstat -ano | findstr :8000
taskkill /PID <PID_NUMBER> /F

netstat -ano | findstr :3000
taskkill /PID <PID_NUMBER> /F
```

4 WINDOWS - CLOUDSHELL LIMITATION

The CloudShell (browser terminal) uses Unix PTY which is not available on native Windows. All other 8 pages work perfectly on Windows.

Feature Compatibility

Feature	Windows Native	Windows + WSL2
Dashboard	Works	Works
Deploy Wizard	Works	Works
Policy Dashboard	Works	Works
Audit Log	Works	Works
Team Management	Works	Works
Drift Detection	Works	Works
Admin Settings	Works	Works
CloudShell Terminal	Won't connect	Works

To Get CloudShell on Windows (WSL2)

```
# PowerShell (Run as Administrator):
wsl --install

# Restart your computer, then open Ubuntu:
cd /mnt/c/path/to/project
chmod +x setup.sh && ./setup.sh
./start.sh
```

5 LOGIN & USER ROLES

Open <http://localhost:3000>. No password needed - just enter a username.

Default Login

Field	Value
Username	Eternal-prithivi
Password	Not required (username lookup only)

Available Roles

Role	Access Level
admin	Full - all pages, CloudShell, settings, user management
devops	Most - deploy, policies, CloudShell, drift detection
developer	Limited - dashboard, policies, audit logs
viewer	Read-only - dashboard and audit only

All usernames are in team-management/teams.yaml

6 TROUBLESHOOTING

WebSocket connection failed in CloudShell

- **Cause:** Backend not running, or websockets package missing
- **Fix:** Use `.venv/bin/uvicorn` (macOS) or `.venv\Scripts\uvicorn.exe` (Windows)
- **Verify:** Open `http://localhost:8000/api/health` - should show `{"status":"ok"}`

Cannot connect to API at localhost:8000

- **Cause:** Backend server is not running
- **Fix:** Start the backend first, then refresh the browser

Module not found errors in Python

- **Cause:** Not using the virtual environment
- **Fix:** Always use `.venv/bin/python` (macOS) or `.venv\Scripts\python.exe` (Windows)

Frontend shows blank page

- **Cause:** Node.js dependencies not installed
- **Fix:** Run: `cd web-ui/frontend && npm install && npm run dev`

'python3' not recognized (Windows)

- **Cause:** Python not added to PATH during installation
- **Fix:** Reinstall Python, check 'Add Python to PATH'. Use 'python' not 'python3'

Port already in use

macOS/Linux:

```
lsof -i :8000 -t | xargs kill -9
lsof -i :3000 -t | xargs kill -9
```

Windows:

```
netstat -ano | findstr :8000
taskkill /PID <number> /F
```

7 QUICK REFERENCE CARD

Action	macOS / Linux	Windows
Setup	./setup.sh	setup.bat
Start	./start.sh	start.bat
Stop	./start.sh --stop	start.bat --stop
Run tests	.venv/bin/python -m pytest tests/	.venv\Scripts\python.exe -m pytest tests/
Python	.venv/bin/python	.venv\Scripts\python.exe
Uvicorn	.venv/bin/uvicorn	.venv\Scripts\uvicorn.exe
pip install	.venv/bin/pip install X	.venv\Scripts\pip.exe install X

8 RUNNING TESTS

macOS / Linux

```
.venv/bin/python -m pytest tests/ -v          # All 308 tests
.venv/bin/python -m pytest tests/unit/test_server.py -v # API only
.venv/bin/python -m pytest tests/ --cov        # With coverage
```

Windows

```
.venv\Scripts\python.exe -m pytest tests/ -v          # All 308 tests
.venv\Scripts\python.exe -m pytest tests\unit\test_server.py -v
.venv\Scripts\python.exe -m pytest tests/ --cov        # With coverage
```